

# A Coverage–Identifiability Calculus for Biosynthetic Program Inference and Design

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companion to “Sound Biosynthetic Design over Iterative Grammars”

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***Status of this draft.** Skeleton + framing prose + every empirical claim wired to its reproduction command. The formal development (definitions, lemmas, theorems, proofs) lives in full in the methods document `complete_miner_method.md` (§16); this manuscript packages it as a standalone paper. Prose set per the author’s delegation; final voice and the headline-adjacent phrasing are the author’s. Two results enter as pre-registration (macrolactonization, lanthipeptide structural datapoint), never as claimed datapoints, until their external inputs exist.*

## Abstract

The companion paper poses fungal-polyketide structure prediction as verifier-grounded inference over latent biosynthetic programs, and shows the field’s accuracy ceiling is a statement about grammar identifiability rather than data volume. Here we extract the family-agnostic object underneath that instance: a **coverage–identifiability calculus**. A sound executor’s three-state verdict partitions any biosynthetic corpus into verified (R), under-observed (U), and out-of-grammar (X) sets, governed by two knobs with opposite monotonicities — an operator library  $\Theta$  (coverage) and an observable set  $O$  (resolution).  $O$ -enrichment moves  $U \rightarrow R$  and can never exit X;  $\Theta$ -enrichment moves  $X \rightarrow R \cup U$  and can demote  $R \rightarrow U$ . The asymmetry makes coverage ( $|X|$ ) and identifiability ( $|U|$ ) distinct degrees of freedom that trade, and identifies the honest monotone progress metric as  $R^{\text{term}}$  (verified under the full operator universe), not the raw verified count. We prove the verdict is provably terminating — every cluster resolves under product, genomic, or trajectory observation, with the within-enzyme product mixture the sole permanent floor — and that the observability quantity  $\kappa_O$  stratifies into a **modality ladder** (product / genomic / trajectory) whose cells are family-invariant. A controllability–modality theorem shows the design difficulty axes (structural vs control edits) *are* the genome vs trajectory modalities, deriving rather than positing the field’s measured cost structure; and the cross-taxa prediction wall is re-read as the genome→trajectory modality crossing. We validate the calculus against the executor on fungal PKS (the partition, both monotonicities, the completeness certificate,  $R^{\text{term}}$ , the controllability duality) and instantiate its combinatorial half on a non-PKS family (lanthipeptide, on a provenanced lacticin-481 core), with the structural datapoint held behind an honest validation-independence gate. Every reported number reproduces from a single documented command.

## 1 Introduction

The companion paper makes a specific claim about fungal polyketide synthases: the product is the image  $y = \text{Exec}_\Theta(z; G)$  of a latent program  $z$ , and the field’s 51–68% accuracy ceiling reflects grammar identifiability, not sample size. That paper’s machinery — a sound deterministic executor, a beam verifier returning the exact consistent set  $Z^*$ , and a three-state reconstructibility verdict —

is built for fungal PKS but is, by construction, not specific to them: only the program generator is family-specific, while the observability and verdict layers are shared.

This paper isolates what is general. We show the three-state verdict, read as a *partition over a corpus*, obeys a small calculus with two control knobs and a single observability quantity, and that this calculus (i) makes precise what “mining a corpus” means and what its honest progress metric is, (ii) certifies its own enumeration completeness, (iii) proves a termination property for the verdict, and (iv) unifies the inference and design halves of the companion paper through a modality refinement of its identifiability–controllability duality. The fungal instance becomes one occupant of a family-invariant structure; a second family (lanthipeptide) is instantiated to test cross-family transfer.

The contribution is a theory layer that travels: any biosynthetic family expressible as a sound executor over a program grammar inherits the calculus, the termination guarantee, and the modality ladder. The fungal results of the companion paper validate it; the lanthipeptide results test it on chemistry the original engine never saw.

## 2 The three-state partition and the $(\Theta, O)$ calculus

(Full development: *complete\_miner\_method.md* §16.1–16.2.)

For a grammar  $G$ , operator library  $\Theta$ , and observable set  $O$ , write  $\kappa_{\Theta, O}(y) = |Z_{\Theta}^*(y; G, O)|$ . Each corpus member falls in exactly one of  $R$  ( $\kappa = 1$ , verified),  $U$  ( $\kappa \geq 2$ , under-observed),  $X$  ( $\kappa = 0$ , out-of-grammar). By Theorem 1 of the companion paper,  $\kappa_{\Theta, O}$  is simultaneously the inference ambiguity and the design freedom at  $y$  — the partition is read identically by predictor and designer.

A granularity the calculus surfaces immediately:  $\kappa_O$  counts *programs* (Theorem 1’s equivalence class), so  $R$  is the program-verified set, distinct from the structure-reachable set. On MIBiG 4.0 fungal PKS this is the reach-9 / verified-8 distinction — nine structures placed, eight to a unique program, the ninth (olivetolic) a sound two-route degeneracy.

**The two knobs, opposite monotonicities.**  $O$ -enrichment refines the observable signature, so  $\kappa$  is non-increasing: it moves  $U \rightarrow R$  and leaves  $X$  invariant (you cannot observe your way out of a coverage gap).  $\Theta$ -enrichment enlarges the program space, so  $\kappa$  is non-decreasing: it moves  $X \rightarrow R \cup U$  and can *demote*  $R \rightarrow U$  (a richer operator set can manufacture a second route to an already-unique structure). Coverage and identifiability are therefore distinct degrees of freedom that trade; the monotone quantities are  $|X|$  (coverage) and, at fixed  $\Theta$ ,  $|U|$  (identifiability) — never  $|R|$ .

**Coverage flow.** The  $\Theta$ -move transition matrix has three theorem-forced zeros ( $R \not\rightarrow X$ ,  $U \not\rightarrow R$ ,  $U \not\rightarrow X$ ); the live cells are  $X \rightarrow R$  (clean coverage),  $X \rightarrow U$  (coverage needing an observable),  $R \rightarrow U$  (ambiguity births). The empirically decisive law: **demotion is driven by operator overlap, not growth** — adding a redundant route demotes, adding disjoint chemistry grows  $R$  cleanly (§6).

## 3 Reach, $R^{\text{term}}$ , and a runnable completeness certificate

(Full development: §16.3–16.4.)

Because  $|R|$  is non-monotone, the honest progress metric is  $R^{\text{term}}$ : cores verified under the full operator universe  $\mathcal{U}$ , not merely the current  $\Theta$ .  $|R^{\text{term}}|$  is monotone non-decreasing in  $\Theta$ ; a core verified only because  $\Theta$  is impoverished (provisional) demotes to  $U$  as  $\Theta \rightarrow \mathcal{U}$ . The reach-9/verified-8 gap is exactly this: olivetolic has  $\kappa(\cdot; \mathcal{U}) \geq 2$ , so it is never terminal —  $|R^{\text{term}}| = 8$ , stable across the declared beam-adequate universe (terminality is universe-relative; §16.4a).

**$\kappa$ -monotonicity is a completeness certificate.** The beam verifier can only under-count ( $\kappa_\beta \leq \kappa$ ), and  $\kappa$  is monotone in  $\Theta$ ; so a reported decrease  $\kappa_\beta(\Theta') < \kappa_\beta(\Theta)$  for  $\Theta \subseteq \Theta'$  *proves* the larger search pruned a valid program. The relative-completeness caveat of the companion paper’s §4.1 thereby becomes a runnable per-core diagnostic — a zero-false-alarm detector of beam inadequacy, complemented by exhaustion as the sufficient certificate. This fired in practice (a methylmalonyl-extended scan inverted  $\kappa$  at  $\beta = 8000$ ; widening to  $5 \times 10^4$  restored monotonicity).

**Validation independence.** Executor correctness is certified by reconstructing a *known* structure to a  $\simeq$ -match — sound only if the known structure is determined independently of the executor’s inputs. A program-derived (e.g. sequence+connectivity) anchor is what the executor consumes, so matching against it is a mirror, not a gate. This is why families with scarce deposited structures are exactly the U-verdict families: the validation wall and the identifiability wall coincide.

## 4 The modality ladder and the controllability–modality duality

(Full development: §16.7, §16.13.)

The observability quantity  $\kappa_O$  stratifies into mutually non-refining **modalities**: *product* (mass, MS/MS, NMR — functions of the structure), *genomic* (domain/sequence/loading-domain — functions of  $\phi(B)$ ), and *trajectory* (per-cycle intermediate, isotope provenance). Product-only observation can refine no finer than the fibres of  $\text{Exec}_\Theta$  (which is non-injective); sub-fibre resolution needs a latent modality.

**Trajectory-completeness.** A program’s provenance-tagged intermediate trajectory determines it uniquely, so every  $\kappa \geq 2$  fibre closes under some modality — there is no permanent  $U_\infty$ ; the within-enzyme product mixture (one sequence, one substrate,  $\geq 2$  products) is the sole permanent floor, and there the mixture certificate *is* the correct output. Hence the three-state verdict is provably terminating: every cluster resolves to  $R^{\text{term}}$ , U (tagged with its cheapest closing modality), or X.

**Controllability–modality (the design half).** A design edit changes a program feature living in the genome or trajectory modality. The wet-lab capability that realizes it equals the modality of that feature: structural edits are genome-class (genome-readable, genome-engineerable — mature combinatorial biosynthesis), control edits are trajectory-class (genome-invisible, the iteration-program frontier). This *derives* the companion paper’s §9 design axes and the *shape* of its super-additive cost (genome edits separable  $\rightarrow$  linear; trajectory edits coupled through one reused manifold  $\rightarrow$  super-additive), rather than positing them. The cross-taxa prediction wall ( $1.00 \rightarrow 0.567$ ) is the genome $\rightarrow$ trajectory modality crossing: bacterial modular control closes at the genome modality (per-module sequence reads the program), fungal iterative control closes only at trajectory.

## 5 Cross-family: family-invariant cells

(Full development: §16.13–16.14.)

The  $(m_{\text{ctrl}}, m_{\text{conf}})$  cells are **family-invariant** — the design-side face of the shared observability/verdict layer; what differs across families is which chemistry occupies which cell. Fungal PKS populate every cell, with cycle-assignment in the (trajectory, trajectory) cell (the genuine hard, product-neutral U). The framework predicts a non-PKS family will occupy the same cells with different chemistry.

**The cells, measured (make design-tag).** Tagging each edit of the fungal 1-edit neighbourhood by its program feature’s modality, *independently* of §9’s structural/control axis label, the two are in conflict in **0 of 192** edits — Theorem 16.13 confirmed corpus-wide, not merely on the 6-MSA neighbourhood. Occupancy: (genome, product) = 46, (genome, latent) = 3 (the olivetolic-class product-neutral starter swaps, genome-read), (trajectory, product) = 69 (every frontier edit mass-confirmable, matching §9.3’s “(0,1) verified by accurate mass”), and (trajectory, trajectory) = **0**. The empty (trajectory, trajectory) cell is decisive: the iteration-grammar U-class is **empirically un-witnessed** in the reachable neighbourhood — confirming, not asserting, the coverage-gating §16.12 pre-registers (it lives beyond the current boundary, in X).

**Lanthipeptide (RiPP), pre-registered and partly validated.** Core-residue identity is genome-modality (read off the ORF). Ring topology is the (trajectory-control, product-confirm) cell — genome-prediction-limited and design-limited (Repka’s barrier), yet product-resolvable (the bridges are bonds). Because lanthionine bridging is mass-neutral, every topology of a core is an isomer:  $\kappa_{\text{mass}} = (\text{number of legal pairings})$ , so the topology-U fibre is *factorial* in ring count — derived, not asserted. The combinatorial half is validated on a real provenanced core (lacticin 481); the structural  $\simeq$ -datapoint is held behind the validation-independence gate (no independent deposited atomic structure exists for the 27-mer), which is itself consistent with the U-verdict.

## 6 Results (each wired to a reproduction command)

All numbers below reproduce from the open code base; commands as named (make targets in the repository) plus the lanthipeptide additions.

| Result                         | Claim                                                                                                                | Reproduce                                                  |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| Partition (fungal PKS)         | R = 8, U = 1 (olivetolic), X = 317; reach 9                                                                          | <code>make reachability</code>                             |
| Monotonicity asymmetry         | starter sweep:  R  9→8, reach 9→9, one demotion; forbidden cells empty                                               | <code>make coverage-sweep</code>                           |
| Overlap-not-growth             | release+PT sweeps: clean X → R, 0 demotions                                                                          | <code>make coverage-sweep</code>                           |
| R <sup>term</sup> monotone     | 8 (robust to +methylmalonyl) → 11 (+PT), clean                                                                       | <code>make coverage-sweep</code>                           |
| Completeness certificate       | $\kappa$ -inversion caught $\beta=8000$ pruning; certified the PT climb                                              | <code>make kappa-check</code>                              |
| Cross-taxa = modality crossing | genome modality: bacterial 60 → 70/118 (+10), 31/58 silent-KR; fungal floor 0.567                                    | <code>make seqhead</code> , <code>make differential</code> |
| Controllability duality        | 69:49 frontier:engineerable = trajectory:genome (0/192 edits in conflict); (0, 1) edits mass-confirmable             | <code>make design-tag</code>                               |
| Lanthipeptide combinatorics    | $\kappa_{\text{mass}} = P(6, 3)=120 / P(4, 3)=24$ on lacticin-481 core; containment of {9–14, 11–25, 18–26}; isomass | <code>make test</code>                                     |

Suite: 181 tests pass (161 prior + 20 lanthipeptide). Provenance for the lanthipeptide core: UniProt P36499, cross-validated against FEBS Lett. 1996 (PMID 8764998) and JACS 2013 (PMC3736828).

## 7 Limitations and the two pre-registered forward edges

The calculus is established without external inputs; what remains is gated and enters as falsifiable pre-registration with turnkey executors behind it.

- **Macrolactonization (the iteration-grammar trajectory-U witness).** Predicts a small but nonzero count of same-product / same-starter / different-cycle fibres among newly-reachable HR macrolactone cores — the first live occupant of the (trajectory, trajectory) cell. Gated on a sound macrolactonization operator (the rung-2/3 “cost-open” ring chemistry).
- **Lanthipeptide structural datapoint.** Predicts ring topology lands in (trajectory-control, product-confirm), MS/MS/NMR-resolvable, genome-prediction-limited. Gated on an independent deposited atomic structure clearing the  $\simeq$ -gate; the scaffold (enumeration + partition harness) is built and suite-green, the structural renderers stubbed behind tripwire tests.

Four live falsifiers attach to the lanthipeptide prediction; the macrolactonization climb is certified by the same  $\kappa$ -monotonicity completeness check. Neither claims a datapoint before its gate clears.

## 8 Conclusion

A sound executor over a biosynthetic program grammar carries more than a predictor: its three-state verdict is a partition obeying a two-knob calculus in which coverage and identifiability are distinct, trading quantities; whose honest progress metric is  $R^{\text{term}}$ ; whose enumeration self-certifies by  $\kappa$ -monotonicity; and whose verdict provably terminates, with the within-enzyme mixture the only permanent floor. The observability quantity stratifies into a family-invariant modality ladder that unifies the inference and design halves of the companion paper — the cross-taxa prediction wall and the design-difficulty axes are one quantity read in two modalities. The fungal instance validates the calculus; lanthipeptide tests it on a second family. What the framework cannot yet claim, it pre-registers, behind a validation-independence gate that ties the limits of what is buildable to the limits of what is independently observable — the same wall, named honestly.

## References

- [1] B. Eilertsen. *Sound Biosynthetic Design over Iterative Grammars: A Realizability Geometry for Fungal Polyketides* (2026a).
- [2] Riedling et al. 2024 (the 51–68% ceiling); Cox 2023 (iterative HR-PKS programming); Repka et al. 2017 (lanthipeptide enzymology); Crawford et al. 2009, Li et al. 2010 (PT-domain regiochemistry); Eng/Tao et al. (ClusterCAD); Zdouc et al. 2025 (MIBiG 4.0); Lin et al. 2023 (ESM-2); Landrum et al. 2025 (RDKit); van den Hooven et al. 1996 (lacticin 481 NMR); Knerr & van der Donk 2013 (lacticin 481 synthesis). (*Full bibliography inherited from the companion paper plus the lanthipeptide sources.*)

*Formal proofs, the full transition-matrix and modality-ladder treatment, all definitions/lemmas/theorems, and the per-result derivations are in `complete_miner_method.md` (§16), which serves as this paper’s methods/supplement until merged.*